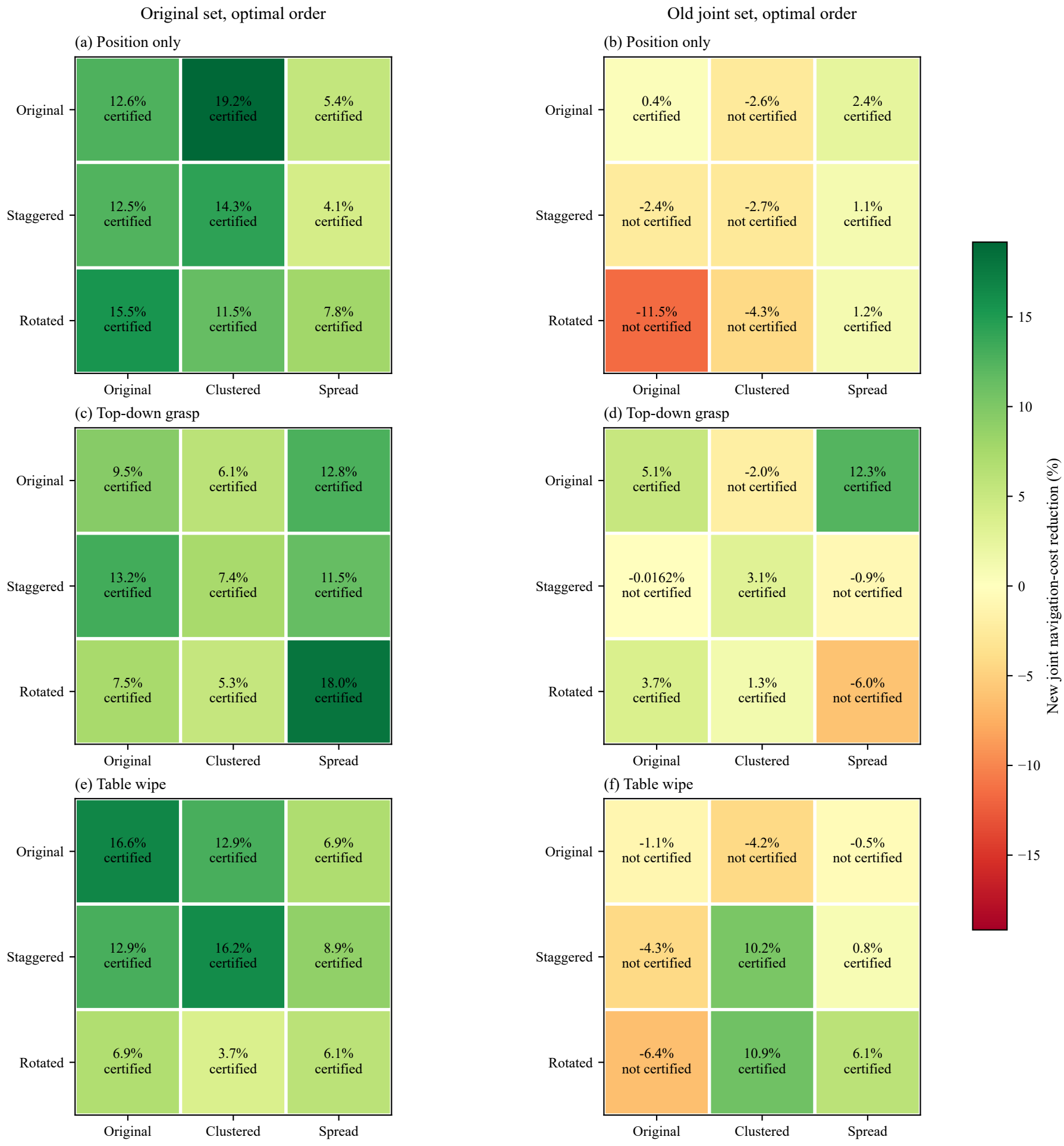
Original set, optimal order
Old joint set, optimal order
(a) Position only
12.6% certified
19.2% certified
5.4% certified
12.5% certified
14.3% certified
4.1% certified
15.5% certified
11.5% certified
7.8% certified
(b) Position only
0.4% certified
-2.6% not certified
2.4% certified
-2.4% not certified
-2.7% not certified
1.1% certified
-11.5% not certified
-4.3% not certified
1.2% certified
(c) Top-down grasp
9.5% certified
6.1% certified
12.8% certified
13.2% certified
7.4% certified
11.5% certified
7.5% certified
5.3% certified
18.0% certified
(d) Top-down grasp
5.1% certified
-2.0% not certified
12.3% certified
-0.0162% not certified
3.1% certified
-0.9% not certified
3.7% certified
1.3% certified
-6.0% not certified
(e) Table wipe
16.6% certified
12.9% certified
6.9% certified
12.9% certified
16.2% certified
8.9% certified
6.9% certified
3.7% certified
6.1% certified
(f) Table wipe
-1.1% not certified
-4.2% not certified
-0.5% not certified
-4.3% not certified
10.2% certified
0.8% certified
-6.4% not certified
10.9% certified
6.1% certified
Original
Staggered
Rotated
Original
Clustered
Spread
New joint navigation-cost reduction (%)
15
10
5
0
−5
−10
−15